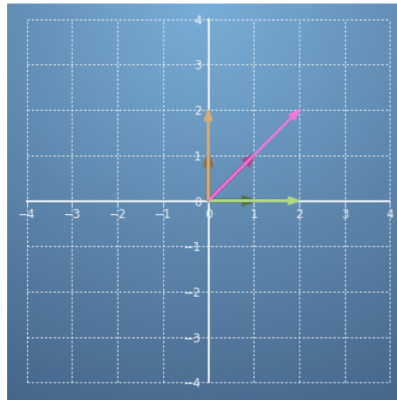


- Recall that for a linear transformation, an eigenvector is a vector which, after applying the transformation, stays in the same span. In the following questions, you will try to geometrically see which vectors of a linear transformation are eigenvectors.

1 point

In the following diagram, the dark green vector is given by $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, the purple vector by $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and the brown vector by $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

The transformation $T = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ is applied, which sends the three vectors to the light green vector $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$, the magenta vector $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ and the orange vector $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$, respectively.



Which of the three original vectors are eigenvectors of the linear transformation T ?

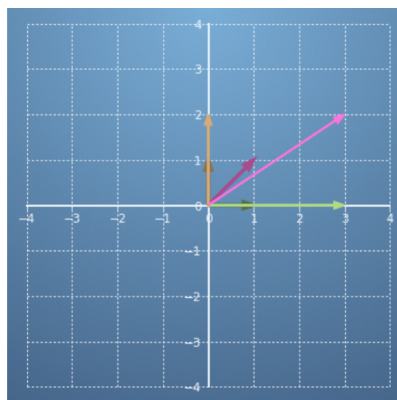
- ☒ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
- ☒ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- ☒ $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
- ☐ None of the above.

- Recall that for a linear transformation, an eigenvector is a vector which, after applying the transformation, stays in the same span. In the following questions, you will try to geometrically see which vectors of a linear transformation are eigenvectors.

1 point

In the following diagram, the dark green vector is given by $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, the purple vector by $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and the brown vector by $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

The transformation $T = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$ is applied, which sends the three vectors to the light green vector $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$, the magenta vector $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ and the orange vector $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$, respectively.



Which of the three original vectors are eigenvectors of the linear transformation T ?

- ☒ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

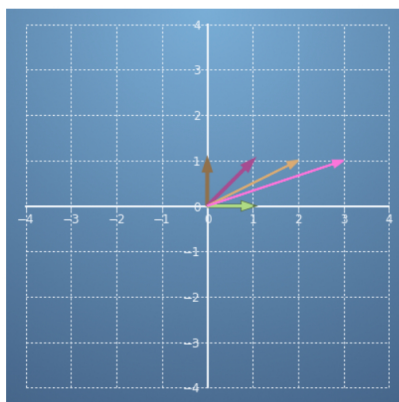
- ☐ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- ☒ $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
- ☐ None of the above.

3. Recall that for a linear transformation, an eigenvector is a vector which, after applying the transformation, stays in the same span. In the following questions, you will try to geometrically see which vectors of a linear transformation are eigenvectors.

1 point

In the following diagram, the dark green vector is given by $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, the purple vector by $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and the brown vector by $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

The transformation $T = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ is applied, which sends the three vectors to the light green vector $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, the magenta vector $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and the orange vector $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, respectively.



Which of the three original vectors are eigenvectors of the linear transformation T ?

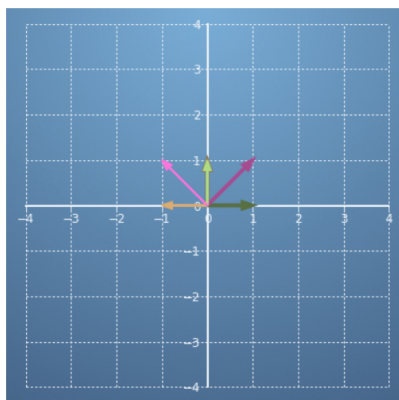
- ☒ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
- ☐ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- ☐ $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
- ☐ None of the above.

4. Recall that for a linear transformation, an eigenvector is a vector which, after applying the transformation, stays in the same span. In the following questions, you will try to geometrically see which vectors of a linear transformation are eigenvectors.

1 point

In the following diagram, the dark green vector is given by $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, the purple vector by $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and the brown vector by $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

The transformation $T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is applied, which sends the three vectors to the light green vector $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, the magenta vector $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and the orange vector $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$, respectively.



Which of the three original vectors are eigenvectors of the linear transformation T ? Select all correct answers.

☐ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

☐ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

☐ $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

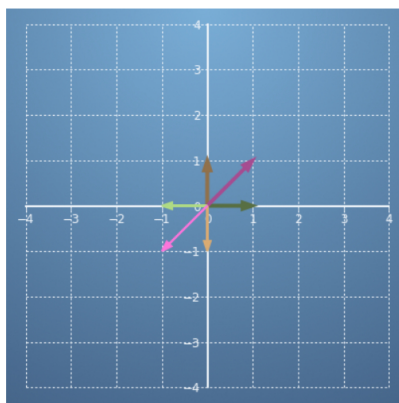
☒ None of the above.

5. Recall that for a linear transformation, an eigenvector is a vector which, after applying the transformation, stays in the same span. In the following questions, you will try to geometrically see which vectors of a linear transformation are eigenvectors.

1 point

In the following diagram, the dark green vector is given by $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, the purple vector by $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and the brown vector by $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

The transformation $T = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ is applied, which sends the three vectors to the light green vector $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$, the magenta vector $\begin{bmatrix} -1 \\ -1 \end{bmatrix}$ and the orange vector $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$, respectively.



Which of the three original vectors are eigenvectors of the linear transformation T ?

☒ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

☒ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

☒ $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

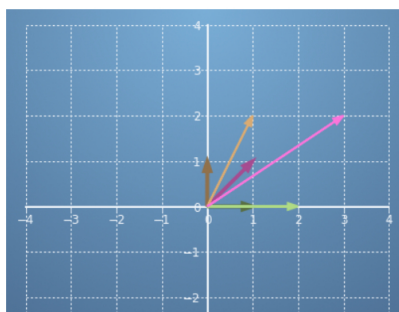
☐ None of the above

6. Recall that for a linear transformation, an eigenvector is a vector which, after applying the transformation, stays in the same span. In the following questions, you will try to geometrically see which vectors of a linear transformation are eigenvectors.

1 point

In the following diagram, the dark green vector is given by $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, the purple vector by $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and the brown vector by $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

The transformation $T = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ is applied, which sends the three vectors to the light green vector $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$, the magenta vector $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and the orange vector $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, respectively.





Which of the three original vectors are eigenvectors of the linear transformation T ?

☒ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

☐ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

☐ $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

☐ None of the above.